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United States Patent Application Publication

20250282365

Kind Code

A1

Publication Date

September 11, 2025

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APPARATUS AND METHOD FOR SETTING FAULT-TOLERANT TIME INTERVAL WHEN VEHICLE DRIVING

Abstract

The present disclosure relates to an apparatus and method for setting a fault-tolerant time interval when vehicle driving, and includes a memory comprising at least one instruction; and at least one processor configured to execute the at least one instruction stored in the memory, wherein the at least one processor is configured to identify whether sensing data is obtained from a sensor provided in a handle of a vehicle when a malfunction is detected in at least one torque sensor provided in driving vehicle, and set a fault-tolerant time interval (FTTI) to increase according to a predetermined reference value in the vehicle when the sensing data is obtained, and is applicable to other embodiments.

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Appl. No.: 18/806143

Filed: August 15, 2024

Foreign Application Priority Data

KR 10-2024-0032610

Mar. 07, 2024

Publication Classification

Int. Cl.: B60W50/02 (20120101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0032610, filed on Mar. 7, 2024, the disclosures of which is incorporated herein by reference in its entirety.

FIELD

[0002] The present disclosure relates to an apparatus and method for setting a fault-tolerant time interval when vehicle driving.

BACKGROUND

[0003] An electric power steering (EPS) system is a system that uses a processor to adjust the power steering characteristics of a handle by controlling the amount of assist for driving a motor. The processor is a control device that receives electrical signals detected by various input sensors of the vehicle and outputs digital control signals to driving various actuators on the output side.

[0004] When the driver operates the handle, the torque sensor detects the driver's handle torque input and transmits it to the processor, and the motor is driven by a command of the processor receiving the handle torque input and assists the steering, so that the EPS system controls the power steering characteristics of the handle. That is, the processor may control the power steering of the handle to become lighter when the vehicle speed is low, and may control the power steering of the handle to become heavier when the vehicle speed is high.

[0005] Recently, Fault-tolerant time interval (FTTI), which is a failure detection time for detecting the failure of the EPS system, has been applied due to the spread of autonomous vehicles. In this case, when the FTTI is set to be large, there is a problem that a malfunction phenomenon generated in the vehicle cannot be detected early, and when the FTTI is set to be low, the malfunction phenomenon generated in the vehicle is sensitively detected, resulting in problems such as warning lights occurring and assist not being possible. In particular, in order to prevent sensitive detection of malfunctions, a sensor with high resolution or a CPU with high response and processing speed must be applied, which inevitably leads to an increase in costs.

[0006] Therefore, there is an emerging need to develop a technology that can set an appropriate FTTI in real time or periodically when a malfunction phenomenon occurs in a vehicle without installing a separate sensor.

SUMMARY

[0007] Embodiments of the present disclosure for solving the conventional problems provide an apparatus and method for setting a fault-tolerant time interval when vehicle driving, which may set an appropriate fault-tolerant time interval (FTTI) when a malfunction occurs in the vehicle using sensing data obtained from a sensor mounted on a handle of the vehicle.

[0008] According to an embodiment of the present disclosure, an apparatus for setting fault-tolerant time interval when vehicle driving includes a memory including at least one instruction; and at least one processor configured to execute the at least one instruction stored in the memory, wherein the at least one processor is configured to identify whether sensing data is obtained from a sensor provided in a handle of a vehicle when a malfunction is detected in at least one torque sensor provided in driving vehicle, and set a fault-tolerant time interval (FTTI) to increase according to a predetermined reference value in the vehicle when the sensing data is obtained.

[0009] In addition, the at least one processor may differentially set the FTTI according to a touch state and a grasp state of the handle identified from the sensing data.

[0010] In addition, the at least one processor may set the FTTI to decrease when the sensing data is

not obtained.

[0011] In addition, the at least one processor may identify an operation result value for sensing data obtained from the at least one torque sensor when the malfunction is not detected in the at least one torque sensor, and identify whether a malfunction is generated in the vehicle based on the operation result value.

[0012] In addition, the at least one processor may identify the operation result value based on the sensing data obtained from the torque sensor using a first algorithm and a second algorithm.

[0013] In addition, the at least one processor may identify that the malfunction is generated in the vehicle when a difference between the operation result value identified by the first algorithm and the operation result value identified by the second algorithm is greater than or equal to a threshold value.

[0014] In addition, the at least one processor may compare each sensing data obtained by the first torque sensor and the second torque sensor provided in the vehicle, and detect whether the at least one torque sensor malfunctions based on whether each sensing data is the same.

[0015] In addition, the at least one processor may include a first processor and a second processor, and the first processor and the second processor apply the first algorithm and the second algorithm, respectively.

[0016] In addition, according to an embodiment of the present disclosure, a method of setting a fault-tolerant time interval when vehicle driving may include detecting, using an electronic apparatus, a malfunction in at least one torque sensor provided in driving vehicle; identifying, using an electronic apparatus, whether sensing data is obtained from a sensor provided in a handle of the vehicle when the malfunction is detected in at least one torque sensor; and setting, using an electronic apparatus, a fault-tolerant time interval (FTTI) to increase according to a predetermined reference value in the vehicle when the sensing data is obtained from a sensor provided in the handle.

[0017] In addition, the setting the FTTI to increase may be differentially setting the FTTI according to a touch state and a grasp state of the handle identified from the sensing data.

[0018] In addition, the method may further include setting, using an electronic apparatus, the FTTI to decrease when the sensing data is not obtained.

[0019] In addition, the method may further include after the detecting whether the at least one torque sensor malfunctions, identifying, using an electronic apparatus, an operation result value for sensing data obtained from the at least one torque sensor when the malfunction is not detected in the at least one torque sensor; and

[0020] identifying, using an electronic apparatus, whether a malfunction is generated in the vehicle based on the operation result value.

[0021] In addition, the identifying the operation result value may be identifying the operation result value based on the sensing data obtained from the torque sensor using a first algorithm and a second algorithm.

[0022] In addition, the identifying whether the malfunction is generated in the vehicle may be identifying that the malfunction is generated in the vehicle when a difference between the operation result value identified by the first algorithm and the operation result value identified by the second algorithm is greater than or equal to a threshold value.

[0023] In addition, the detecting whether the at least one torque sensor malfunctions may include comparing each sensing data obtained by the first torque sensor and the second torque sensor provided in the vehicle; and detecting whether the at least one torque sensor malfunctions based on whether each sensing data is the same.

[0024] As described above, the apparatus and method for setting a fault-tolerant time interval when vehicle driving according to the present disclosure have the effect of setting an appropriate fault-tolerant time interval (FTTI) when the occurrence of a malfunction phenomenon of the vehicle by using sensing data obtained from a sensor mounted on a handle of the vehicle, so that the FTTI can

be set in real time or periodically without installing a separate sensor, and there is a cost saving effect because there is no need to install the separate sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 is a diagram showing a main configuration of a system for setting a fault-tolerant time interval according to an embodiment of the present disclosure.

[0026] FIG. 2 is a flowchart for describing a method of setting a fault-tolerant time interval according to an embodiment of the present disclosure.

[0027] FIG. 3 is a diagram illustrating a main configuration of a system for setting a fault-tolerant time interval according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0028] Hereinafter, preferred embodiments according to the present disclosure will be described in detail with reference to the accompanying drawings. The detailed description that will be disclosed below together with the accompanying drawings is intended to explain exemplary embodiments of the present disclosure, and is not intended to represent the only embodiments in which the present disclosure can be practiced. In the drawings, parts irrelevant to the description may be omitted to clearly explain the present disclosure, and the same reference numerals may be used for the same or similar components throughout the specification.

[0029] FIG. 1 is a diagram showing a main configuration of a system for setting a fault-tolerant time interval according to an embodiment of the present disclosure.

[0030] Referring to FIG. 1, the system **10** according to the present disclosure may include a hands on detection (HOD) sensor **100**, an electronic apparatus **150**, and a steering wheel **200**. In addition, FIG. 1 may be applied when the vehicle's autonomous driving level is level **1** and level **2**.

[0031] The HOD sensor **100** is a sensor provided in the handle of the vehicle, and may be a sensor that may detect a microcurrent generated by a driver's hand when the driver touches or grasps the handle. The HOD sensor **100** provides the detected sensing data (hereinafter, referred to as third sensing data) to the electronic apparatus **150**.

[0032] When a malfunction is detected in at least one torque sensor provided in the driving vehicle, the electronic apparatus **150** identifies whether the third sensing data is obtained from the HOD sensor **100** provided in the handle of the vehicle, and sets a fault-tolerant time interval to decrease or increase depending on whether the third sensing data is obtained. To this end, the electronic apparatus **150** may include a first torque sensor **151**, a second torque sensor **152**, a memory **153**, a processor **154**, and a motor **155**. The electronic apparatus **150** may be an electric power steering (EPS) system that adjusts the power steering characteristics of the handle by controlling the amount of assist for driving the motor **155**.

[0033] The first torque sensor **151** and the second torque sensor **152** are sensors that obtain first sensing data and second sensing data for the steering torque of the driving vehicle, respectively. The first torque sensor **151** and the second torque sensor **152** obtain the sensing data for the steering torque when vehicle driving and provide the sensing data to the processor **154**. In this case, the first torque sensor **151** may be a main sensor, and the second torque sensor **152** may be a sub-sensor.

[0034] The memory **153** may store a program for controlling the electronic apparatus **150**. In particular, the memory **153** may store a detection algorithm that may identify whether to detect a malfunction in at least one of the first torque sensor **151** and the second torque sensor **152** based on the first sensing data and the second sensing data received from the first torque sensor **151** and the second torque sensor **152**.

[0035] The memory **153** may store a first algorithm and a second algorithm that may identify

whether to detect a malfunction in the electronic apparatus **150** by applying the first sensing data and the second sensing data. In this case, the first algorithm may be an algorithm for calculating an actual operation result value related to driving of the vehicle based on the first sensing data and the second sensing data obtained according to driving of the vehicle, and the second algorithm may be an algorithm for identifying consistency of the actual operation result value calculated by the first algorithm. In addition, the memory **153** may store a reference value that is a maximum value when setting the FTTI to increase.

[0036] The processor **154** identifies whether the vehicle starts driving. When it is identified that the vehicle starts driving, the processor **154** identifies whether the malfunction is detected in at least one torque sensor among the first torque sensor **151** and the second torque sensor **152** based on the first sensing data and the second sensing data received from the first torque sensor **151** and the second torque sensor **152**. More specifically, the processor **154** applies the first sensing data and the second sensing data to the detection algorithm stored in the memory **153** to compare them. The processor **154** may identify that when the two sensing data differ by a threshold value or more, the malfunction is detected in the torque sensor. At this time, the processor **154** may detect hardware damage such as a device or circuit damage of the torque sensor as a torque sensor malfunction.

[0037] Conversely, the processor **154** identifies that when the two sensing data are similar or identical to less than the threshold, no malfunction is detected in the torque sensor. When it is identified that no malfunction is detected in the torque sensor, the processor **154** applies the first sensing data and the second sensing data to the first algorithm and the second algorithm stored in the memory **153**. The processor **154** may apply the first sensing data and the second sensing data to the first algorithm to calculate the operation result value, and may apply the first sensing data and the second sensing data to the second algorithm to calculate the operation result value. In this case, the first algorithm and the second algorithm are algorithms for pre-processing the first sensing data and the second sensing data, and may mean an algorithm that calculates an operation result value by performing operations such as converting the first sensing data and the second sensing data into a torque value.

[0038] The processor **154** compares the operation result values calculated from each of the first algorithm and the second algorithm. When the difference between the two operation result values is greater than or equal to the threshold value or the difference between the operation result values is greater than or equal to the threshold value and is maintained for a threshold time or longer, the processor **154** identifies the differences as a software error such as a code defect or operation delay of the first or second algorithm, and determines that the malfunction is detected in the electronic apparatus **150**, that is, the EPS system. Conversely, the processor **154** may identify that when the two operation result values are similar or identical to those of the threshold value, a separate malfunction has not been detected in the EPS system.

[0039] When it is identified that a malfunction is detected in the torque sensor or the EPS system, the processor **154** identifies whether third sensing data is obtained from the hands on detection (HOD) sensor **100** provided in the handle of the vehicle. When the third sensing data is not obtained, the processor **154** identifies that the driver is not touching or grasping the handle of the vehicle and sets the fault-tolerant time interval (FTTI) to decrease.

[0040] Conversely, when the third sensing data is obtained from the HOD sensor **100**, the processor **154** identifies that the driver is touching or grasping the handle of the vehicle and sets FTTI to increase. At this time, the processor **154** may determine whether the driver is touching or grasping the handle of the vehicle based on the third sensing data and set the FTTI differently depending on the state. For example, when the driver is grasping the handle of the vehicle, the FTTI may be set to increase by the reference value stored in the memory **153**, and when the driver is touching the handle of the vehicle, the FTTI may be set to increase, but lower than the reference value.

[0041] The processor **154** may perform the corresponding function based on the set FTTI. For example, the processor **154** sets FTTI to decrease when the third sensing data is not obtained while

the malfunction is detected in the torque sensor or EPS system of the vehicle. The processor **154** may display a warning light that is generated whenever the reduced FTTI arrives and a message notifying that steering assist is not possible and that the driver must directly control the steering. In this way, when the FTTI is reduced and set short, situations in which the warning lights and the steering assist are not possible are sensitively detected, so the driver may be encouraged to participate in real-time steering control to ensure safety when vehicle driving.

[0042] In addition, when the third sensing data is obtained in a state where the malfunction is detected in the torque sensor or EPS system of the vehicle, the processor **154** sets FTTI to increase. The processor **154** may prevent sensitive detection by setting the FTTI to increase and display the warning light that is generated whenever the FTTI arrives and the message notifying that steering assist is not possible and the driver must directly control the steering. As described above, when the third sensing data is obtained while the malfunction is detected in the torque sensor or the EPS system, the processor **154** may determine that the driver is controlling the steering by using the handle, and may encourage the driver to participate in the steering control when it is determined that the driver's steering control is necessary at the time of arrival of FTTI.

[0043] In addition, the processor **154** adjusts the amount of assist based on the set FTTI and provides a current corresponding to the adjusted amount of assist to the motor **155**.

[0044] The motor **155** is controlled according to a current corresponding to the amount of assist adjusted based on the FTTI set by the processor **154**.

[0045] The steering wheel **200** may adjust the power steering characteristics of the handle according to the control of the motor **155**.

[0046] FIG. **2** is a flowchart for describing a method of setting a fault-tolerant time interval according to an embodiment of the present disclosure.

[0047] Referring to FIG. **2**, in step **S201**, the processor **154** identifies whether the vehicle starts driving. As a result of the step **S201**, when it is identified that the vehicle starts driving, the processor **154** performs step **S203**, and when the driving start is not identified, the processor **154** continuously identifies whether the vehicle starts driving.

[0048] In step **S203**, the processor **154** identifies whether the malfunction is detected in at least one torque sensor among the first torque sensor **151** and the second torque sensor **152** based on the first sensing data and the second sensing data received from the first torque sensor **151** and the second torque sensor **152**. More specifically, the processor **154** applies the first sensing data and the second sensing data to the detection algorithm to compare them. The processor **154** identifies that when the two sensing data differ by a threshold value or more, the malfunction is detected in the torque sensor and performs step **S207**. At this time, the processor **154** may detect hardware damage such as a device or circuit damage of the torque sensor as a torque sensor malfunction.

[0049] Conversely, the processor **154** identifies that when the two sensing data are similar or identical to less than the threshold, no malfunction is detected in the torque sensor, and performs step **S205**.

[0050] In step **S205**, the processor **154** may apply the first sensing data and the second sensing data to the first algorithm to calculate the operation result value, and may apply the first sensing data and the second sensing data to the second algorithm to calculate the operation result value. In this case, the first algorithm and the second algorithm are algorithms for pre-processing the first sensing data and the second sensing data, and may mean an algorithm that calculates an operation result value by performing operations such as converting the first sensing data and the second sensing data into a torque value.

[0051] The processor **154** compares the calculated results of each of the first algorithm and the second algorithm. When the difference between the two operation result values is greater than or equal to the threshold value or the difference between the operation result values is greater than or equal to the threshold value and is maintained for a threshold time or longer, the processor **154** identifies the differences as a software error such as a code defect or operation delay of the first or

second algorithm, and determines that the malfunction is detected in the electronic apparatus **150**, that is, the EPS system. When it is identified that the malfunction is detected in the EPS system, the processor **154** performs step **S207**. On the contrary, when the two operation result values are similar or identical to those of the threshold value, it is identified that a separate malfunction has not been detected in the EPS system.

[0052] In step **207**, the processor **154** identifies whether third sensing data is obtained from the hands on detection (HOD) sensor **100** provided in the handle of the vehicle. When it is identified that the third sensing data is obtained as a result of step **S207**, the processor **154** performs step **S209**, and when it is identified that the third sensing data is not obtained, the processor **154** performs step **S211**. In step **S211**, since the third sensing data is not obtained, the processor **154** identifies that the driver is not touching or grasping the handle of the vehicle and sets the fault-tolerant time interval (FTTI) to decrease, and performs step **S213**. For example, the processor **154** may set the FTTI to 20 ms.

[0053] On the other hand, if it is identified that the third sensing data is obtained, the processor **154** may perform the step **S213** after setting the FTTI to increase by identifying that the driver is touching or grasping the handle of the vehicle in step **S209**. At this time, the processor **154** may determine whether the driver is touching or grasping the handle of the vehicle based on the third sensing data and set the FTTI differently depending on the state. For example, when the driver is grasping the handle of the vehicle, the FTTI may be set to increase by the reference value stored in the memory **153**, for example, to be set to 70 ms, and when the driver is touching the handle of the vehicle, the FTTI may be set to increase, for example, to be set to 50 ms lower than 70 ms.

[0054] In step **S213**, the processor **154** may perform the corresponding function based on the set FTTI. For example, the processor **154** sets FTTI to decrease when the third sensing data is not obtained while the malfunction is detected in the torque sensor or EPS system of the vehicle. The processor **154** may display a warning light that is generated whenever the reduced FTTI arrives and a message notifying that steering assist is not possible and that the driver must directly control the steering. In this way, when the FTTI is reduced and set short, situations in which the warning lights and the steering assist are not possible are sensitively detected, so the driver may be encouraged to participate in real-time steering control to ensure safety when the vehicle driving.

[0055] In addition, when the third sensing data is obtained in a state where the malfunction is detected in the torque sensor or EPS system of the vehicle, the processor **154** sets FTTI to increase. The processor **154** may prevent sensitive detection by setting the FTTI to increase and display the warning light that is generated whenever the FTTI arrives and the message notifying that steering assist is not possible and the driver must directly control the steering. As described above, when the third sensing data is obtained while the malfunction is detected in the torque sensor or the EPS system, the processor **154** may determine that the driver is controlling the steering by using the handle, and may encourage the driver to participate in the steering control when it is determined that the driver's steering control is necessary at the time of arrival of FTTI.

[0056] In addition, in step **S213**, the processor **154** adjusts the amount of assist based on the set FTTI and provides a current corresponding to the adjusted amount of assist to the motor **155**. The motor **155** is controlled according to the current, and The steering wheel **200** adjusts the power steering characteristics of the handle according to the control of the motor **155**.

[0057] In step **S215**, the processor **154** may terminate the corresponding process when the end of the driving of the vehicle is detected, and return to step **S203** when the end of the driving of the vehicle is not detected, and repeatedly perform steps **S203** to **S213**.

[0058] FIG. **3** is a diagram illustrating a main configuration of a system for setting a fault-tolerant time interval according to another embodiment of the present disclosure.

[0059] Referring to FIG. **3**, the system **30** according to another embodiment of the present disclosure may include a hands on detection (HOD) sensor **300**, an electronic apparatus **350**, and a steering wheel **400** as shown in FIG. **1**. In addition, FIG. **3** may be applied when the vehicle's

autonomous driving level is level 3.

[0060] Since the HOD sensor **300** performs the same function as the HOD sensor **100** shown in FIG. 1, a detailed description thereof will be omitted.

[0061] When a malfunction is detected in at least one torque sensor provided in the driving vehicle, the electronic apparatus **350** identifies whether the third sensing data is obtained from the HOD sensor **300** provided in the handle of the vehicle, and when the third sensing data is obtained, sets a fault-tolerant time interval to increase according to a predetermined reference value in the vehicle. To this end, the electronic apparatus **350** may include a first torque sensor **351**, a second torque sensor **352**, a memory **353**, a processor **354**, a first motor **355a**, and a second motor **355b**.

[0062] At this time, the first torque sensor **351**, the second torque sensor **352**, and the memory **153** perform the same function as the first torque sensor **151**, the second torque sensor **152**, and the memory **153** shown in FIG. 1, and thus a detailed description thereof will be omitted.

[0063] The processor **354** may include a first processor **354a** and a second processor **354b**, and the first processor **354a** and the second processor **354b** may perform separately the functions of the processor **154** shown in FIG. 1. In this case, the first processor **354a** may perform the role of the main processor.

[0064] For example, the first processor **354a** receives first sensing data and second sensing data from the first torque sensor **351** and the second torque sensor **352**, and the second processor **354b** receives the first sensing data and the second sensing data from the first torque sensor **351** and the second torque sensor **352**. The first processor **354a** compares the first sensing data and the second sensing data by applying the first sensing data and the second sensing data obtained by the first torque sensor **351** and the second torque sensor **352** to a detection algorithm. The first processor **354a** may identify that when the two sensing data differ by a threshold or more, the malfunction is detected in the torque sensor. The first processor **354a** may detect hardware damage such as a device or circuit damage of the torque sensor as a torque sensor malfunction.

[0065] Conversely, the first processor **354a** identifies that when the two sensing data are similar or identical to less than the threshold, no malfunction is detected in the torque sensor. When it is identified that no malfunction is detected in the torque sensor, the first processor **354a** calculates the operation result values for the first sensing data and the second sensing data using a first algorithm stored in the memory **353**. The second processor **354b** calculates the calculation result values for the first sensing data and the second sensing data using a second algorithm stored in the memory **353**.

[0066] The first processor **354a** may receive the operation result value calculated by the second processor **354b** and detect whether the EPS system is malfunction. More specifically, the first processor **354a** compares the operation result value calculated by the first processor **354a** with the operation result value calculated by the second processor **354b**. When the difference between the two operation result values is greater than or equal to the threshold value or the difference between the operation result values is greater than or equal to the threshold value and is maintained for a threshold time or longer, the first processor **354a** identifies the differences as a software error such as a code defect or operation delay of the first or second algorithm, and determines that the malfunction is detected in the electronic apparatus **350**, that is, the EPS system. Conversely, the first processor **354a** may identify that when the two operation result values are similar or identical to those of the threshold value, a separate malfunction has not been detected in the EPS system.

[0067] When the first processor **354a** identifies that a malfunction is detected in the torque sensor or the EPS system, the first processor **354a** identifies whether the third sensing data is obtained from the HOD sensor **300** provided in the handle of the vehicle. When the third sensing data is not obtained, the first processor **354a** identifies that the driver is not touching or grasping the handle of the vehicle and sets the fault-tolerant time interval (FTTI) to decrease.

[0068] Conversely, when the third sensing data is obtained from the HOD sensor **300**, the first processor **354a** identifies that the driver is touching or grasping the handle of the vehicle and sets

FTTI to increase. At this time, the first processor **354a** may determine whether the driver is touching or grasping the handle of the vehicle based on the third sensing data and set the FTTI differently depending on the state. For example, when the driver is grasping the handle of the vehicle, the FTTI may be set to increase by the reference value stored in the memory **353**, and when the driver is touching the handle of the vehicle, the FTTI may be set to increase, but lower than the reference value.

[0069] The first motor **355a** and the second motor **355b** are controlled according to a current corresponding to the amount of assist adjusted based on the FTTI set by the first processor **354a**. At this time, the first motor **355a** may serve as a main motor, and when an error occurs in the first motor **355a**, the second motor **355b** may serve as the first motor **355a** instead.

[0070] The steering wheel **400** may adjust the power steering characteristics of the handle according to the control of the first motor **355a** or the second motor **355b**.

[0071] The embodiments of the present disclosure disclosed in this specification and the drawings are merely presented to easily explain the technical contents of the present disclosure and to help understand the present disclosure, and the scope of the present disclosure is not intended to be limited. Therefore, it should be interpreted that all changes or modified forms derived based on the technical spirit of the present disclosure are included in the scope of the present disclosure in addition to the embodiments disclosed herein.

Claims

1. An electronic apparatus, comprising: a memory comprising at least one instruction; and at least one processor configured to execute the at least one instruction stored in the memory, wherein the at least one processor is configured to: identify whether sensing data is obtained from a sensor provided in a handle of a vehicle when a malfunction is detected in at least one torque sensor provided in driving vehicle, and set a fault-tolerant time interval (FTTI) to increase according to a predetermined reference value in the vehicle when the sensing data is obtained.
2. The electronic apparatus of claim 1, wherein the at least one processor is configured to differentially set the FTTI according to a touch state and a grasp state of the handle identified from the sensing data.
3. The electronic apparatus of claim 1, wherein the at least one processor is configured to set the FTTI to decrease when the sensing data is not obtained.
4. The electronic apparatus of claim 1, wherein the at least one processor is configured to: identify an operation result value for sensing data obtained from the at least one torque sensor when the malfunction is not detected in the at least one torque sensor, and identify whether a malfunction is generated in the vehicle based on the operation result value.
5. The electronic apparatus of claim 4, wherein the at least one processor is configured to identify the operation result value based on the sensing data obtained from the at least one torque sensor using a first algorithm and a second algorithm.
6. The electronic apparatus of claim 5, wherein the at least one processor is configured to identify that the malfunction is generated in the vehicle when a difference between the operation result value identified by the first algorithm and the operation result value identified by the second algorithm is greater than or equal to a threshold value.
7. The electronic apparatus of claim 1, wherein the at least one processor is configured to: compare each sensing data obtained by a first torque sensor and a second torque sensor provided in the vehicle, and detect whether the at least one torque sensor malfunctions based on whether each sensing data is the same.
8. The electronic apparatus of claim 5, wherein the at least one processor comprises a first processor and a second processor, and the first processor and the second processor apply the first algorithm and the second algorithm, respectively.

9. A method for setting a fault-tolerant time interval, comprising: detecting, using an electronic apparatus, a malfunction in at least one torque sensor provided in driving vehicle; identifying, using an electronic apparatus, whether sensing data is obtained from a sensor provided in a handle of the vehicle when the malfunction is detected in at least one torque sensor; and setting, using an electronic apparatus, a fault-tolerant time interval (FTTI) to increase according to a predetermined reference value in the vehicle when the sensing data is obtained from a sensor provided in the handle.

10. The method of claim 9, wherein the setting the FTTI to increase is differentially setting the FTTI according to a touch state and a grasp state of the handle identified from the sensing data.

11. The method of claim 9, further comprising: setting, using an electronic apparatus, the FTTI to decrease when the sensing data is not obtained.

12. The method of claim 9, further comprising: after the detecting whether the at least one torque sensor malfunctions, identifying, using an electronic apparatus, an operation result value for sensing data obtained from the at least one torque sensor when the malfunction is not detected in the at least one torque sensor; and identifying, using an electronic apparatus, whether a malfunction is generated in the vehicle based on the operation result value.

13. The method of claim 12, wherein the identifying the operation result value is identifying the operation result value based on the sensing data obtained from the at least one torque sensor using a first algorithm and a second algorithm.

14. The method of claim 13, wherein the identifying whether the malfunction is generated in the vehicle is identifying that the malfunction is generated in the vehicle when a difference between the operation result value identified by the first algorithm and the operation result value identified by the second algorithm is greater than or equal to a threshold value.

15. The method of claim 9, wherein the detecting whether the at least one torque sensor malfunctions comprising: comparing each sensing data obtained by a first torque sensor and a second torque sensor provided in the vehicle; and detecting whether the at least one torque sensor malfunctions based on whether each sensing data is the same.
